THE STACK OF RELATIVELY PERFECT COMPLEXES IS LIMIT PRESERVING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-complexes (canonical tag 0DLE). cited at statement line 5614. Standing situation: X -> B and the relatively perfect K.
- Definition artin-definition-limit-preserving (canonical tag 07XL). cited at statement line 5617. The property being established: limit preservation.
- Remark spaces-limits-remark-limit-preserving (canonical tag 05N0). cited at proof line 5636. Bottom arrow equivalence; reduce to limits over B.
- Lemma spaces-more-morphisms-lemma-descend-relatively-perfect (canonical tag 0DKW). cited at proof lines 5642-5643. T-perfect objects = colimit of T_i-perfect objects.
- Lemma lemma-complexes (canonical tag 0DLD). cited at proof lines 5650 and 5653. Condition (2) being descended.
- Lemma lemma-complexes-open-neg-exts-vanishing (canonical tag 0DLC). cited at proof line 5655. The open W of negative-Ext vanishing for E_i.
- Lemma limits-lemma-limit-contained-in-constructible (canonical tag 05F4). cited at proof line 5660. T -> T_i factors through W at a finite stage i'.

Source reference. Original definition 0DLE: sources/stacks-chow/tags.txt, source lines 17044–17044. Cited at statement line 5614: the lemma is phrased in this situation; load-bearing.

Source reference. Original definition 07XL: sources/stacks-chow/tags.txt, source lines 10081–10081. Cited at statement line 5617: the property being established; load-bearing.

Source reference. Original remark 05N0: sources/stacks-chow/tags.txt, source lines 7289–7289. Cited at proof line 5636: the bottom arrow of the diagram is an equivalence since B -> S is locally of finite presentation; load-bearing.

Source reference. Original lemma 0DKW: sources/stacks-chow/tags.txt, source lines 17026–17026. Cited at proof lines 5642-5643: the category of T-perfect objects is the colimit; load-bearing.

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0DLD: sources/stacks-chow/tags.txt, source lines 17043–17043. Cited at proof lines 5650 and 5653: condition (2) of that lemma; load-bearing twice.

Source reference. Original lemma 0DLC: sources/stacks-chow/tags.txt, source lines 17042–17042. Cited at proof line 5655: the open W constructed for E_i; load-bearing.

Source reference. Original lemma 05F4: sources/stacks-chow/tags.txt, source lines 7013–7013. Cited at proof line 5660: T -> T_i factors through W at some finite stage i'; load-bearing.

Lemma 0.2. *In Situation 0DLE assume that $B \to S$ is locally of finite presentation. Then $p : \mathit{Complexes}_{X/B} \to (\mathit{Sch}/S)_{fppf}$ is limit preserving (Artin's Axioms, Definition 07XL).*

Proof. Write $B(T)$ for the discrete category whose objects are the S-morphisms $T \to B$. Let $T = \lim T_i$ be a filtered limit of affine schemes over S. Assigning to an object (T, h, E) of $\mathit{Complexes}_{X/B,T}$ the object h of $B(T)$ gives us a commutative diagram of fibre categories

$$\begin{array}{ccc} \operatorname{colim} \mathit{Complexes}_{X/B,T_i} & \longrightarrow & \mathit{Complexes}_{X/B,T} \\ \downarrow & & \downarrow \\ \operatorname{colim} B(T_i) & \longrightarrow & B(T) \end{array}$$

We have to show the top horizontal arrow is an equivalence. Since we have assume that B is locally of finite presentation over S we see from Limits of Spaces, Remark 05N0 that the bottom horizontal arrow is an equivalence. This means that we may assume $T = \lim T_i$ be a filtered limit of affine schemes over B. Denote $g_i : T_i \to B$ and $g : T \to B$ the corresponding morphisms. Set $X_i = T_i \times_{g_i,B} X$ and $X_T = T \times_{g,B} X$. Observe that $X_T = \operatorname{colim} X_i$. By More on Morphisms of Spaces, Lemma 0DKW the category of T-perfect objects of $D(\mathcal{O}_{X_T})$ is the colimit of the categories of T_i-perfect objects of $D(\mathcal{O}_{X_{T_i}})$. Thus all we have to prove is that given an T_i-perfect object E_i of $D(\mathcal{O}_{X_{T_i}})$ such that the derived pullback E of E_i to X_T satisfies condition (2) of Lemma 0DLD, then after increasing i we have that E_i satisfies condition (2) of Lemma 0DLD. Let $W \subset |T_i|$ be the open constructed in Lemma 0DLC for E_i and E_i. By assumption on E we find that $T \to T_i$ factors through T. Hence there is an $i' \geq i$ such that $T_{i'} \to T_i$ factors through W, see Limits, Lemma 05F4 Then i' works by construction of W. $\square$